BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024 • Paper Code: PR–XII (A-I-A2) (24)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** PR–XII (A-I-A2) (24)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. Which of the given acids is found in rancid butter?

(A) $C_2H_4O_2$ (B) $C_8H_6O_2$

(C) $C_4H_8O_2$ (D) $C_6H_{10}O_2$

Q2. Which of the given is the strongest acid?

(A) Ethanol (B) Acetic acid

(C) Chloroacetic acid (D) Flouro acetic acid

Q3. Steroids belong to the family of:

(A) Protein (B) Lipid

(C) Enzyme (D) Carbohydrates

Q4. Which of the given is condensation polymer?

(A) PVC (B) Polythene

(C) Nylon (D) Rubber

Q5. Photochemical smog is primarily caused by:

(A) CO (B) CO_2

(C) O_2 (D) NO_2

Q6. UV ranges from:

(A) 400-800 nm
(B) 800-1200 nm
(C) 1200-1611 nm
(D) 10-400 nm

Q7. Which of the given halide is most reactive towards nucleophilic substitution reaction?

(A) C2H5F
(B) C2H6Cl
(C) C2H5Br
(D) C2H5I

Q8. The end product of reaction of Grignard reagent with ethyl acetate is:

(A) Primary alcohol
(B) Secondary alcohol
(C) Tertiary alcohol
(D) None of these

Q9. Oxidation of secondary alcohols gives;

(A) Carboxylic acid
(B) Ether
(C) Aldehyde
(D) Ketone

Q10. Phenol containing 5% of water is known as:

(A) Carbolic acid
(B) Picric acid
(C) Ethanoic acid
(D) Methanoic acid

Q11. The carbon of the carbonyl group is:

(A) Sp^3-hybridized
(B) Sp^2-hybridized
(C) Sp-hybridized
(D) Un-hybridized

Q12. Which of the given compounds can react with ammonical silver nitrate solution to form silver mirror:

(A) Acetone
(B) Ethanol
(C) Ether
(D) Ethanal

Q13. Which of the given metal sulphates is almost insoluble in water?

(A) Na2SO4
(B) K2SO4
(C) CaSO4
(D) BaSO4

Q14. The coordination number of iron in $[Fe(CN)6]^{-4}$ is:

(A) 2
(B) 3
(C) 4
(D) 6

Q15. Pyridine belongs to which class of organic compounds?

(A) Heterocyclic
(B) Homocyclic
(C) Alicyclic
(D) Acyclic

Q16. Which of the given represeots the functional group of amides:

(A) -COOR
(B) -NH2
(C) -CONH2
(D) -C≡N

Q17. Which one of the given compounds exhibit geometric isomerism?

(A) 1-butane
(B) 2-butane
(C) 2-butyne
(D) 2-Methyl propane

Q18. Which one of the given is O/P direction but deactivates benzene ring?

(A) -Cl
(B) -NH2
(C) -OH
(D) -OCH3

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q2. Answer any TEN parts. Each part carries FOUR marks.

1. How Beryllium differs from the other members of its own group?
2. Explain the catalytic behavior of transition elements?
3. Differentiate between partial and total synthesis?
4. Describe any two methods of preparation of alkynes?
5. What is meant by chirality? Give an example.
6. Why tertiary carbocation is more stable than secondary and primary carbocation?
7. What is Lucas Test?
8. Explain briefly which tests can be used for identification of aldehydes from ketones in laboratory?
9. Why chloro acetic acid is stronger acid than acetic acid?
10. Differentiate between DNA and RNA.
11. What are the uses of PVC and Nylon?
12. Why is acid rain considered as a threat to historical monuments?
13. Why odern methjod of analysis is superior over classical methods of analysis?

SECTION-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 27

Note: Note: Attempt any THREE questions. All questions carries equal marks.

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Q3.

(a) Give IUPAC names to the following structures. (i) (CH3)4C (ii) benzene ring with COCH3 substituent (iii) CH2(OH)–CH2(OH) (iv) CH3–C≡C–COOH (v) CH3–CH(O–CH2–CH3)–CH2–CH3

(b) What is meant by inert pair effect? Explain inert pair effect in formation of ionic and covalent bond.

Q4.

(a) Starting from alkene how would you prepare the following compounds? (i) Vicinal dihalide (ii) Alcohol (iii) Aldehyde (iv) Epoxide

(b) Discuss the mechanism to SN1 and SN2 reactions.

Q5.

(a) Write the four different methods of preparation of Phenols.

(b) Define enzymes. Describe the role of enzymes in digestion of carbohydrates and proteins.

Q6.

(a) What is greenhouse effect? How is it causing global warming?

(b) Explain the construction and working of mass spectrometery?